

Scikit-learn and TensorFlow are a few open source machine learning tools.

Cloud Foundry, Cloudify and OpenShift are a few open source PaaS platforms.

The key considerations when choosing an open source cloud platform are listed below:

- Deployment flexibility
- Ease of operations
- Choice of application stacks
- Language, database and framework support
- Scaling capabilities
- QoS
- Tooling for development and operations
- How well it meets the needs of the business

An open source cloud is designed and developed using open source technologies and software such as:

- An open source operating system
- An open source mobile OS
- Open source application servers
- Open source middleware
- Open source database
- Open source analytics database
- Open source software development frameworks
- Open source workflow and business applications
- Open source monitoring tools
- Open source virtualisation stack (hypervisor, virtualisation management)



Figure 4: Open source options from cloud service providers

- Open source security solutions
- The following are some of the key open source cloud platforms.

OpenStack: This is open source software for creating private and public clouds, built and disseminated by a large and democratic community of developers, in collaboration with users. OpenStack is mostly deployed as Infrastructure-as-a-Service (IaaS), whereby virtual servers and other resources are made available to customers. The software platform consists of interrelated components that control diverse, multi-vendor hardware pools of processing, storage and networking resources throughout a data centre.

Cloud Foundry: This is an open Platform-as-a-Service (PaaS), which provides a choice of clouds, developer frameworks and application services. Cloud Foundry makes it faster and easier to build, test, deploy and scale applications. It has different distributions. A popular one is Pivotal, from IBM.

OpenShift: This is Red Hat's cloud computing PaaS offering. It is an application platform in the cloud, where app developers and teams can build, test, deploy and run their applications.

Cloudify: Cloudify was developed and designed on the principles of openness to power the IT transformation revolution. It enables organisations to design, build and deliver various business applications and network services. The

latest version of Cloudify is 4.3, and it incorporates enhanced features like advanced security, control and true self-service. Cloudify 4.3 introduced a totally new concept for container orchestration with Kubernetes (Kubernetes Cloud Native Orchestration).

WSO2: This is a Platform-as-a-Service (PaaS) framework from the Apache community. It provides the elastic scalability feature for any type of service, using the underlying infra cloud. WSO2 has a microservices based architecture that supports various services. It fosters agility and flexibility with open source.

Cloud Stack: This is an open source software platform designed to manage the cloud computing environment. It is an Infrastructure-as-a-Service (IaaS) cloud computing platform. Various service providers use Cloud Stack to offer public, private and hybrid cloud services.

By adopting open source, enterprises become FAST (flat, agile, streamlined, and tech-savvy). Open source helps to drive digital transformation in enterprises, helping them realise benefits like better business agility, improved customer experiences and enhanced quality of service. Open source based next-generation technologies enable the enterprise to move from transactions based services to relationship based services.

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